retropharyngeal or upper neck nodal recurrence (n=14), skull base recurrence (n=7), and pharyngeal mucosal recurrence (n=5). All patients were immobilized using an in-house, frameless custom cushion-mask-bite-block system and underwent real-time image guidance during each fraction. Planning target volume margin was typically 2 to 3 mm with >95% coverage of prescribed dose and delivered using volumetric modulated arc radiation therapy (VMAT). Treatment-related adverse events were documented according to Common Terminology Criteria for Adverse Events version 4.0. The Kaplain-Meier method was used to estimate clinical endpoints.

Results: Twenty-one patients (81%) had squamous cell carcinoma. All patients received 5 fractions delivered every other day. Twenty-three patients (88%) received a prescribed dose of 45 Gy, 2 patients (8%) received 40 Gy, and 1 patient received 47.5 Gy. Twenty-two patients (86%) received concurrent weekly cetuximab. With a median follow-up time of 6.2 months (range 3.1-20.7 months), the 6-month overall survival, disease-free survival, and locoregional control rates were 79%, 73%, and 91%, respectively. Five patients failed in the head and neck. One had persistent disease in-field after tongue base treatment, 1 recurred in the soft tissue adjacent (<1 cm) to the treated skull base, 1 recurred in an inferior adjacent nodal station after isolated neck treatment, and 2 failed in the ipsilateral neck after oropharynx treatment. No local failures were observed in 9 patients treated for isolated retropharyngeal recurrence (median follow-up of 7.8 months; range 4.2-19.4 months). There were 3 deaths, all due to metastatic disease. There were 5 acute grade 1 adverse events (3 odynophagia, 1 mucositis, and 1 dysgeusia), 2 grade 2 events (1 mucositis and 1 odynophagia) that required intermittent nonnarcotic oral analgesics, and no acute grade 3 toxicity. Five patients (23%) developed grade 1-2 cetuximab-related folliculitis. One patient developed edema of his oropharyngeal flap 10 days after treatment of the bilateral retropharyngeal nodes to 47.5 Gy.

Conclusion: Linac-based SABR utilizing VMAT for head and neck reirradiation appears safe, tolerable, and effective. Longer follow-up is needed to assess late treatment toxicity and tumor control durability.

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Reirradiation for Recurrent and New Primary Head and Neck Cancer: A Single-Institutional Report

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Purpose/Objective(s): Reirradiation with curative intent for recurrent head and neck cancer is clinically challenging. We conducted a retrospective review of our institutional experience to evaluate the feasibility and toxicity of head and neck reirradiation for recurrent or second primary head and neck cancer.

Materials/Methods: Clinical outcomes of 80 patients who underwent reirradiation for recurrent or second primary head and neck cancer were retrospectively reviewed and analyzed. Reirradiation was defined as any overlap between the treatment fields of previous radiation and retreatment. Disease upon reirradiation was classified as either primary tumor site recurrence or neck recurrence. Late toxicity was evaluated and categorized using the Radiation Therapy Oncology Group grading criteria. Longitudinal estimates of survival were calculated using the Kaplan-Meier method. Univariate analysis was performed using Cox proportional hazard and logistic regression to determine predictors of outcomes and severe late toxicity.

Results: From November 1998 to January 2015, 69 of 80 patients (median age 56 [range 26-84], 26 male, 43 female) who underwent reirradiation at our institution were evaluable. Intensity modulated radiation therapy (IMRT) was used in 63 patients. The median follow-up for patients alive at last follow-up was 11 months (range 2 to 142 months). The most common site of recurrence or second primary disease was the neck (n=43, 62%) and the primary tumor site (n=26, 38%). Twenty-two patients underwent salvage surgery upon recurrence before reirradiaton, and 42 received concurrent systemic therapy. Median dose of initial radiation therapy was 68 Gy (range 9 to 136 Gy). With a median interval from the initial radiation therapy of 42 months (range 2 to 322 months), the median reirradiation dose for patients treated with conventional fractionation was 60 Gy (range 14 to 70 Gy). The median tumor volume was 48.95 mL (range 0.7 to 219.64 mL). The 2-year overall survival and progression-free survival rates were 54% and 33%, respectively. The 2-year locoregional and distant control rates were 39% and 65%, respectively. Patients who underwent salvage surgery prior to radiation treatment had significantly better locoregional control, progression-free survival, and overall survival rates ($P<.05$). Severe (grade 3+) late complications were observed in 21 patients (30%). No significant prognostic factors for severe late toxicity were found on univariate analysis.

Conclusion: Our results showed similar locoregional control rates to historical controls for patients undergoing reirradiation of recurrent and second primary head and neck cancer. The observed rates of severe late toxicity were significant. Further studies are necessary to optimize locoregional control while reducing treatment-related morbidity.

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Evaluation of Weekly Paclitaxel, Carboplatin, and Cetuximab in Head and Neck Cancer Patients With Incurable Disease

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Purpose/Objective(s): Weekly paclitaxel, carboplatin, and cetuximab (PCC) has been found to be efficacious and well tolerated in patients with squamous cell carcinoma of the head and neck (SCCHN) with good performance status (PS) when used as induction chemotherapy. Use of PCC in incurable SCCHN in patients with poor PS or in a noninduction setting is an area that warrants further evaluation. Current recommendations for incurable disease consist of a platinum-based regimen with fluorouracil and cetuximab. Studied in patients with PS of 0 to 1, the fluorouracil-based regimens were associated with significant toxicities. Therefore, weekly PCC may offer an appealing, less toxic alternative for incurable patients with poor PS.

Materials/Methods: This retrospective analysis evaluated 41 patients with very advanced or metastatic head and neck cancer who had received PCC (paclitaxel 80 mg/m^2, carboplatin AUC 2, and a cetuximab 400 mg/m^2 loading dose, followed by 250 mg/m^2 weekly) for up to 6 cycles between April 2008 and September 2014. Maximal response achieved and progression-free survival (PFS), as well as dose intensity and adverse effects, were evaluated.

Results: Of the 41 patients evaluated, baseline PS ranged as follows: PS of 2 (41%), PS of 1 (54%), and PS of 0 (5%). Patients received 2 to 6 cycles, averaging 4 cycles. Thirty-one patients (76%) required treatment to be held, delayed, or dose reduced, most commonly for hematologic toxicities. Grades 3/4 neutropenia occurred in 16 patients (39%), grades 1/2 neutropenia in 12 patients (29%), with grades 3/4 thrombocytopenia in 1 patient (2%) and grades 1/2 thrombocytopenia in 2 patients (4%). No patients developed febrile neutropenia or required hospitalization due to treatment. Partial radiographic response occurred in 15 patients (37%), complete radiographic response in 2 patients (5%), stable disease in 14 patients (34%), and progression in 8 patients (20%). PFS ranged from 1.6 to 45 months, with a median duration of 4.6 months, and median overall survival of 5.25 months.

Conclusion: Analysis indicates that use of weekly PCC appears to be an effective and well-tolerated treatment option for patients with incurable squamous cell carcinoma of the head and neck, specifically with PS of 0 to 2.

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Reirradiation With Simultaneously Integrated Boost (SIB) in Patients With Local Recurrence of Squamous Cell Carcinoma of the Head and Neck: Own Experience

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Purpose/Objective(s): Locoregional recurrence is a major cause of death in patients with squamous cell carcinoma of the head and neck (HNSCC). At the moment, there are no clear recommendations and standards regarding the timing, total doses, and dose tolerance of normal tissues to re-exposure. Based on limited studies on the reirradiation with high total doses, we evaluated the tolerability of high-dose reirradiation with simultaneous integrated boost.

Materials/Methods: Fourteen patients with histologically confirmed locoregional recurrence of HNSCC received reirradiation. Median time after primary radiation therapy course was 60 months. The treatment volumes and total doses were formed as follows: gross tumor volume (primary lesion and involved lymph nodes, delineated on computed tomography [CT], magnetic resonance imaging, and 18F-fluorodeoxyglucose positron emission tomography/CT) + clinical target volume (0.5-1.0 cm) + planning target volume (PTV; 0.3-0.5 cm) was treated to the total dose equivalent to 66 to 70 Gy of conventional fractionation, the upper neck (if indicated, levels I-III + PTV 0.5 cm) to 60 Gy, the lower neck (if indicated, levels IV-V + PTV 0.5 cm) − equivalent to 50 Gy. Single doses to these volumes were 2.14 to 2.21 Gy, 2.0 Gy, and 1.8 Gy, respectively. Radiation treatment was once a day, 5 days a week, 6 weeks long (30 fractions). A treatment planning system was used (intensity modulated radiation therapy [IMRT]), and patients were treated with 2 linear accelerator models. According to the literature, in a year after primary irradiation, almost complete recovery of normal tissue tolerances is observed. Tolerances of the eye, lens, optic nerves and chiasm, brain stem, spinal cord, parotid gland, intact mucosa of the mouth and pharynx were not exceeded. Patient positioning accuracy was controlled by kV-imaging daily and cone beam CT weekly.

Results: Three of 14 patients received the full course of radiation therapy without a break. Radiation toxicity manifested with grade 2 oral and pharyngeal mucositis and grade 2 radiation epidermitis. After 1 month, almost complete relief of radiation mucositis and dermatitis was observed. One patient took a break of 7 days after the 25th fraction due to the development of grade 3 mucositis and grade 3 dysphagia.

Conclusion: Using the technique of SIB with IMRT during curative reirradiation of recurrent HNSCC is available with maintaining satisfactory tolerability.

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Outcomes of Patients With Recurrent and Metastatic Squamous Cell Carcinoma of the Oropharynx

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Purpose/Objective(s): Oropharyngeal cancer is increasing in incidence in North America. There is little data to advise patients on their outcomes if they present with metastatic disease or recurrent disease after primary treatment. The objective of our study was to report the outcomes of these patients from a large retrospective database.

Materials/Methods: This study selected all stage III-IV nonmetastatic oropharyngeal squamous cell cancer patients treated with curative intent and metastatic patients treated with palliative intent in our province between 2006 and 2012. Institutional ethics board approval was granted. Patient data was extrapolated from a prospective registry and retrospective review of medical records. Kaplan-Meier estimates of overall survival (OS) and 95% confidence interval (CI) were obtained for recurrent and metastatic patients. Logistic regression was used to explore the association between patient, tumor, and treatment factors for the outcomes of these patients.

Results: Four hundred thirty-three patients were identified from our registry. Of the 433 patients, 12 had metastatic disease on initial presentation and 76 were identified as recurring after initial primary treatment with surgery or radiation therapy, for a total group of 86 patients. Patients with recurrent disease had a median survival time of 8 months and a 1-year OS rate of 41% after being diagnosed with recurrence. Patients with metastatic disease at presentation had a median survival time of 9.6 months and a 1-year OS rate of 17%. Only 5 out of 76 (6.6%) recurrences were successfully salvaged. Thirty percent of patients received palliative chemotherapy, and 40% received palliative radiation therapy. Receiving more than 1 cycle of palliative chemotherapy was significant on univariate analysis (HR 1.67, $P = .03$) for survival, but not on multivariate analysis. Age, comorbidity index, p16 status, and smoking status were not significant in univariate or multivariate analysis.

Conclusion: Patients with recurrent or metastatic oropharyngeal cancer have a poor prognosis. There may be a benefit to palliative chemotherapy, although the percentage of patients receiving chemotherapy was small. More effective strategies are needed to improve the salvage rate of recurrent patients.

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Disease-Free Interval After Primary Chemoradiation Therapy Can Help Predict the Utility of Salvage Surgery in Patients With Squamous Cell Carcinoma of the Larynx

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Purpose/Objective(s): Each year, more than 12,000 people in the United States are diagnosed with cancer of the larynx, and nearly 3700 die from the disease. The 2 primary treatment options for patients with advanced disease are chemoradiation (CRT) or surgery. The results of the Veterans Affairs Laryngeal Study Group demonstrated equivalent 2-year overall survival between these 2 approaches. Due to this, definitive therapy for locally advanced laryngeal cancer has shifted to an organ-preservation approach with surgery increasingly being reserved for salvage; however, approximately 25% of the patients initially undergoing CRT will eventually require salvage total laryngectomy due to recurrent disease or an incompetent larynx. Patients that require salvage surgery have the possibility of long-term survival, but there is a subset of patients that do poorly,